

ECOCIVIX AI

11 — Testing & QA Strategy

TEAM VELOCITY • HackYatra × EDM 3.0 • PRE-HACKATHON PLANNING

Purpose: Define continuous and final QA for ECOCIVIX, with emphasis on the actual judge-facing demo path.

Status: PRE-HACKATHON DRAFT / PLANNING SPECIFICATION

Approval authority: Garv Nain

QA Philosophy

- Testing is continuous, not a final ceremony.
- Every meaningful change should be verified before integration.

Build Checks

- Frontend build; backend startup; dependency consistency; environment configuration; lint/type/syntax checks where available.

Authentication Tests

- Signup/login; invalid credentials; session persistence; logout; citizen/admin access separation; unauthorized admin requests.

Issue Tests

- Create valid issue; required-field validation; invalid input; retrieval; filtering; updates; lifecycle transitions; unauthorized access; missing issue handling.

AI Tests

- Valid structured response; malformed JSON; missing fields; invalid enums; provider timeout; provider failure; fallback behavior; no secret leakage; AI result persistence.

Database Tests

- Create/read/update relationships; constraints; role data; issue timeline; assignment integrity; duplicate/conflict handling where relevant.

Frontend Tests

- Navigation; forms; loading; empty; error; success; responsive layout; keyboard interaction; accessible labels; mobile reporting flow.

Integration Test — Golden Path

- Citizen login → report issue → submit → AI analysis → issue saved → admin sees issue → admin reviews AI → assigns → updates status → citizen sees update → issue resolved/closed → community intelligence reflects real data.

Security QA

- No secrets in source control; server-side authorization; safe input validation; upload validation; appropriate database policies; no private data leakage.

Regression

- After major integration changes rerun the golden path and critical smoke tests.

Performance

- Test initial load, report submission, AI latency, dashboard rendering, map behavior and 3D fallback on realistic hardware.

Demo Readiness

- Use a clean account/data setup; confirm real data; verify production environment; verify AI provider; test network failure fallback where feasible.

Bug Severity

- P0: blocks demo/core functionality. P1: major feature broken. P2: important but workaround exists. P3: cosmetic/minor issue.

Release Gate

- No unresolved P0; no unresolved P1 on the judge-facing path; core data persistence verified; authentication verified; deployment verified; full demo rehearsed.